

Open Decisions Before the Definitive Experiment

The software can already run a multi-prompt benchmark. The definitive study is blocked mainly by decisions about what will be sampled, how prompts will be compared, and which Judge evidence is valid. These choices must be made before seeing the definitive results.

1. Choose the Model

Current position: Gemini profiles have been used for smoke tests, but the benchmark model is not selected.

Decision needed: provider, exact model/version, generation parameters, maximum output, request pacing, retry policy, and reasoning configuration when applicable.

Why first: model behavior, cost, token accounting, and feasible sample size depend on this choice.

2. Freeze the Candidate Prompts

Current position: one control and five paper-derived translation prompts are implemented.

Decision needed: confirm the final candidate set and freeze every prompt file. Completed runs identify prompts by ID, text, and hash, so later edits must create new prompt versions.

3. Decide How a Prompt Wins

Current position: reports preserve initial translation quality, final repaired quality, Judge outcomes, and repair dependence.

Decision needed: define the primary metric, tie-breakers, and smallest difference worth detecting. Decide whether selection prioritizes initial code, final code, or a declared combination.

Why before the run: changing the criterion after seeing results would allow the data to determine both the rule and the winner.

4. Define the Samples

Current position: the population builder selects one accepted C program for each of 3,265 eligible problems. Selecting a capped two or three programs per problem is still under consideration.

Decision needed:

- programs selected per problem;
- prompt-selection sample size;
- final-evaluation sample size;
- confidence or precision target;
- split and sampling seeds;
- stratification, exclusions, and deterministic reserve problems.

Prompt-selection and final-evaluation sets must be disjoint by problem.

5. Freeze Judge Eligibility and Coverage

Current position: execution and reporting work. Exact description-hash mapping produced 1,763 AIZU matches, and mapped-suite prefetch retrieved 1,738 complete suites with 33,301 cases. The remaining

population-wide check is accepted-C baseline validation. The earlier 1,487-suite corpus used incorrect numeric problem associations and is not experiment evidence. All 1,391 AtCoder problems still lack equivalent system-suite coverage.

Decision needed:

- whether every newly mapped and downloaded suite receives an accepted-C baseline pass before sampling;
- whether Judge eligibility is defined before sampling or checked during each experiment run;
- unavailable-suite replacement policy;
- handling or exclusion of special judges and non-unique outputs;
- whether AtCoder problems are excluded or reported with different coverage.

An exactly mapped and downloaded suite is a candidate, not automatically a valid experiment unit. A problem becomes Judge-backed only after an accepted C reference passes locally. Any exclusion or deterministic reserve policy must be fixed before translation outcomes are observed.

6. Define Any Manual Inspection Protocol

Current position: the workbook supports automatic analysis and navigation to source, attempts, reports, and agent interactions. It does not contain reviewer-entry, blinding, scoring, or adjudication sheets.

Decision needed: only if manual coding will contribute research evidence, define what analysts inspect, the rubric, assignment, capacity, and treatment of disagreements. This protocol does not need to live in the automatic analysis workbook.

Improvements That Do Not Block the Design

- Token usage is extracted from both LangChain usage metadata and the OpenAI-compatible response metadata fallback. It must still be verified with the final benchmark provider because providers may omit usage information.
- Visible-suite review is model-based and cannot prove constraints absent from the C source. The implemented bounded review policy and disabled-suite counts must be reported as limitations.
- Workbook presentation can continue to improve as long as automatic field meanings remain stable. Any separate manual coding protocol must be frozen before it produces research evidence.
- Additional logs should be added only when they provide evidence not already available in structured artifacts.

Recommended Order of Work

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choose model
-> freeze prompts and winning criterion
-> define samples and holdout
-> freeze Judge eligibility and choose suite acquisition mode
-> define manual inspection only if it will produce research evidence
-> run a small end-to-end pilot
-> freeze code, data, configuration, and analysis
-> run prompt selection
-> freeze one prompt
-> run final evaluation
```